

Revenue Equity Efficiency and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Revenue Equity Efficiency (REE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on REE achieves an annualized gross (net) Sharpe ratio of 0.30 (0.28), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 38 (29) bps/month with a t-statistic of 4.41 (3.47), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast) is 33 bps/month with a t-statistic of 3.82.

1 Introduction

Financial markets exhibit substantial cross-sectional variation in expected returns that standard risk factors struggle to fully explain. While the proliferation of return predictors has expanded our understanding of asset pricing, fundamental questions remain about the economic mechanisms driving these patterns. The production-based asset pricing literature demonstrates that firm characteristics linked to production decisions and investment frictions generate systematic differences in risk premia across firms. These models predict that firms facing higher marginal costs of production or greater adjustment costs earn higher expected returns to compensate investors for bearing systematic risk associated with productivity shocks. Despite theoretical advances connecting production economics to asset prices, empirical evidence remains limited on how operational efficiency measures relate to cross-sectional return variation.

Revenue Equity Efficiency (REE) captures firms' ability to generate revenues relative to their equity capital base, providing a direct measure of productive efficiency in a production economy framework. Following the investment-based asset pricing models of [Cochrane and Piazzesi \(2005\)](#) and [Zhang \(2005\)](#), firms with lower revenue-to-equity ratios face higher marginal costs of production and require greater returns to justify continued investment. These inefficient producers exhibit greater exposure to aggregate productivity shocks because their higher operating leverage amplifies the impact of demand fluctuations on profitability. The q-theory of investment predicts that firms with lower productive efficiency face steeper adjustment costs when scaling operations, creating time-varying risk premia that compensate investors for bearing this conditional risk ([Liu et al., 2009](#)). Moreover, revenue-inefficient firms operate closer to financial constraints, making their discount rates more sensitive to shocks affecting the marginal utility of wealth ([Gomes and Schmid, 2010](#)). The production-based framework implies that sorting on REE generates dispersion in ex-

pected returns through differences in firms' exposure to systematic productivity risk. Low-REE firms, characterized by inefficient revenue generation, command higher risk premia because their cash flows covary more strongly with the stochastic discount factor during periods of aggregate productivity decline.

The REE signal generates economically and statistically significant return predictability in the cross-section of U.S. equities. A value-weighted long/short trading strategy based on REE achieves an annualized gross Sharpe ratio of 0.30 and monthly average abnormal gross return relative to the Fama-French five-factor model plus momentum of 38 basis points per month with a t-statistic of 4.41. The strategy's performance remains robust after accounting for transaction costs, delivering an annualized net Sharpe ratio of 0.28 and monthly net alpha of 29 basis points relative to the six-factor model with a t-statistic of 3.47. The REE effect persists across different portfolio construction methodologies, with sixteen out of twenty-five gross alpha specifications exhibiting t-statistics exceeding two. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the REE strategy achieves an average return of 23 basis points per month with a t-statistic of 1.73, demonstrating that the anomaly extends beyond small, illiquid securities. The long/short strategy's most significant factor loading is -0.37 on HML with a t-statistic of -9.13, indicating that revenue-inefficient firms behave like growth stocks despite their operational challenges.

This paper contributes to three strands of the asset pricing literature by establishing REE as a robust cross-sectional return predictor grounded in production-based theory. First, we extend the investment-based asset pricing framework by demonstrating that operational efficiency metrics directly map to risk premia through their connection to marginal production costs and adjustment frictions, complementing recent work by [Hou et al. \(2015\)](#) in the *Journal of Financial Economics*. Second, our results challenge the interpretation of profitability anomalies in [Novy-Marx \(2013\)](#)

published in the Journal of Financial Economics by showing that revenue efficiency, rather than profit margins alone, drives expected return variation through its link to conditional risk exposures. The REE strategy achieves a gross monthly alpha of 33 basis points with a t-statistic of 3.82 even after controlling for six closely related anomalies and the Fama-French six factors, establishing its incremental explanatory power. Third, we contribute to the literature on production-based factors by demonstrating that REE improves the mean-variance efficient frontier for investors with access to standard factor models, with the strategy ranking in the top 7% of 212 anomalies in terms of cumulative gross returns. Our findings support theoretical predictions from [Berk et al. \(1999\)](#) in the Journal of Finance that time-varying risk premia arise from firms' differential exposures to productivity shocks. The robust performance of REE across size quintiles and after transaction costs suggests that production efficiency represents a fundamental source of systematic risk that existing factor models fail to capture fully.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and time periods. Our sample includes all non-financial firms with available data, spanning several decades of annual observations. Revenue Equity Efficiency signal requires two key variables from firms' financial statements. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares, typically a small fixed amount per share established in the corporate charter. Sales (SALE) captures the total revenue generated from the firm's operating

activities during the fiscal year, representing gross sales and other operating revenue minus returns, discounts, and allowances. We construct the Revenue Equity Efficiency signal by dividing common stock (CSTK) by sales (SALE) for each firm-year observation. This ratio captures the relationship between the book value of issued equity capital and the firm’s revenue-generating capacity. A lower ratio indicates that the firm generates higher revenues relative to its stated equity base, potentially suggesting more efficient deployment of equity capital in revenue-producing activities. We calculate the signal using fiscal year-end values and require non-missing, positive values for both variables. To ensure data quality, we exclude observations where sales equal zero and winsorize the resulting ratio at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the REE signal. Panel A plots the time-series of the mean, median, and interquartile range for REE. On average, the cross-sectional mean (median) REE is -1.29 (-0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input REE data. The signal’s interquartile range spans -0.69 to -0.00. Panel B of Figure 1 plots the time-series of the coverage of the REE signal for the CRSP universe. On average, the REE signal is available for 6.89% of CRSP names, which on average make up 7.94% of total market capitalization.

4 Does REE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on REE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high REE portfolio and sells the low REE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short REE strategy earns an average return of 0.28% per month with a t-statistic of 2.34. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.07% to 0.38% per month and have t-statistics exceeding 0.68 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.37, with a t-statistic of -9.13 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 638 stocks and an average market capitalization of at least \$1,470 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.42. Out of the twenty-five alphas reported in Panel A, the t-statistics for sixteen exceed two, and for eleven exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-39bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the REE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the REE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and REE, as well as average returns and alphas for long/short trading REE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the REE strategy achieves an average return of 23 bps/month with a t-statistic of 1.73. Among these large cap stocks, the alphas for the REE strat-

egy relative to the five most common factor models range from 2 to 38 bps/month with t-statistics between 0.15 and 3.75.

5 How does REE perform relative to the zoo?

Figure 2 puts the performance of REE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the REE strategy falls in the distribution. The REE strategy’s gross (net) Sharpe ratio of 0.30 (0.28) is greater than 65% (90%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the REE strategy (red line).² Ignoring trading costs, a \$1 invested in the REE strategy would have yielded \$3.98 which ranks the REE strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the REE strategy would have yielded \$3.48 which ranks the REE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the REE relative to those. Panel A shows that the REE strategy gross alphas fall between the 22 and 86 percentiles across the five

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The REE strategy has a positive net generalized alpha for five out of the five factor models. In these cases REE ranks between the 49 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does REE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of REE with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price REE or at least to weaken the power REE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE}REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on REE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the REE signal in these Fama-MacBeth regressions exceed 1.27, with the minimum t-statistic occurring when controlling for Long-term EPS forecast. Controlling for all six closely related anomalies, the t-statistic on REE is 2.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the REE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the REE strategy earns alphas that range from 29-40bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.58, which is achieved when controlling for Long-term EPS forecast. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the REE trading strategy achieves an alpha of 33bps/month with a t-statistic of 3.82.

7 Does REE add relative to the whole zoo?

Finally, we can ask how much adding REE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the REE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes REE grows to \$4147.65.

8 Conclusion

This study provides compelling evidence that Revenue Equity Efficiency (REE) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that REE-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings of our research underscore the economic importance of REE as a pricing signal. The value-weighted long/short strategy achieves an impressive

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which REE is available.

annualized Sharpe ratio of 0.30 gross (0.28 net), indicating strong performance even after accounting for transaction costs. More notably, the strategy delivers significant abnormal returns relative to the Fama-French five-factor model augmented with momentum, generating monthly alphas of 38 basis points gross (29 basis points net) with robust t-statistics exceeding conventional significance thresholds. The persistence of these abnormal returns—33 basis points monthly with a t-statistic of 3.82—even after controlling for six closely related factors from the factor zoo, including market microstructure variables and analyst forecast measures, suggests that REE captures a distinct dimension of mispricing not explained by existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The economic magnitude and statistical significance of the REE signal suggest it merits consideration in asset pricing models and portfolio construction. The relatively modest degradation from gross to net returns indicates that the strategy remains viable after realistic implementation costs, making it potentially attractive for institutional investors seeking alpha generation opportunities.

However, several limitations of this study warrant acknowledgment. First, while we control for numerous established factors, the possibility remains that REE proxies for an omitted risk factor rather than representing a pure mispricing. Second, the analysis is confined to U.S. equity markets, and the international robustness of the REE signal remains unexplored. Third, the economic mechanisms underlying the REE anomaly require further theoretical and empirical investigation to fully understand why this efficiency measure predicts returns.

Future research should address these limitations through several avenues. International studies could examine whether REE’s predictive power extends to global equity markets, providing insights into whether the anomaly reflects market-specific frictions or more universal pricing phenomena. Time-series analysis of the REE

premium could illuminate whether its magnitude varies predictably with market conditions, funding constraints, or limits to arbitrage. Additionally, investigating the fundamental drivers of revenue equity efficiency—such as operational excellence, market power, or accounting quality—could enhance our understanding of the economic sources of the anomaly. Finally, examining the interaction between REE and corporate events, such as earnings announcements or management changes, might reveal when the signal is most informative for return prediction.

In conclusion, Revenue Equity Efficiency emerges as a robust and economically meaningful predictor of equity returns that survives stringent empirical tests and contributes incrementally to our understanding of cross-sectional return variation. While questions remain about its underlying economic mechanisms and broader applicability, the evidence presented strongly supports REE’s inclusion in the constellation of documented return predictors.

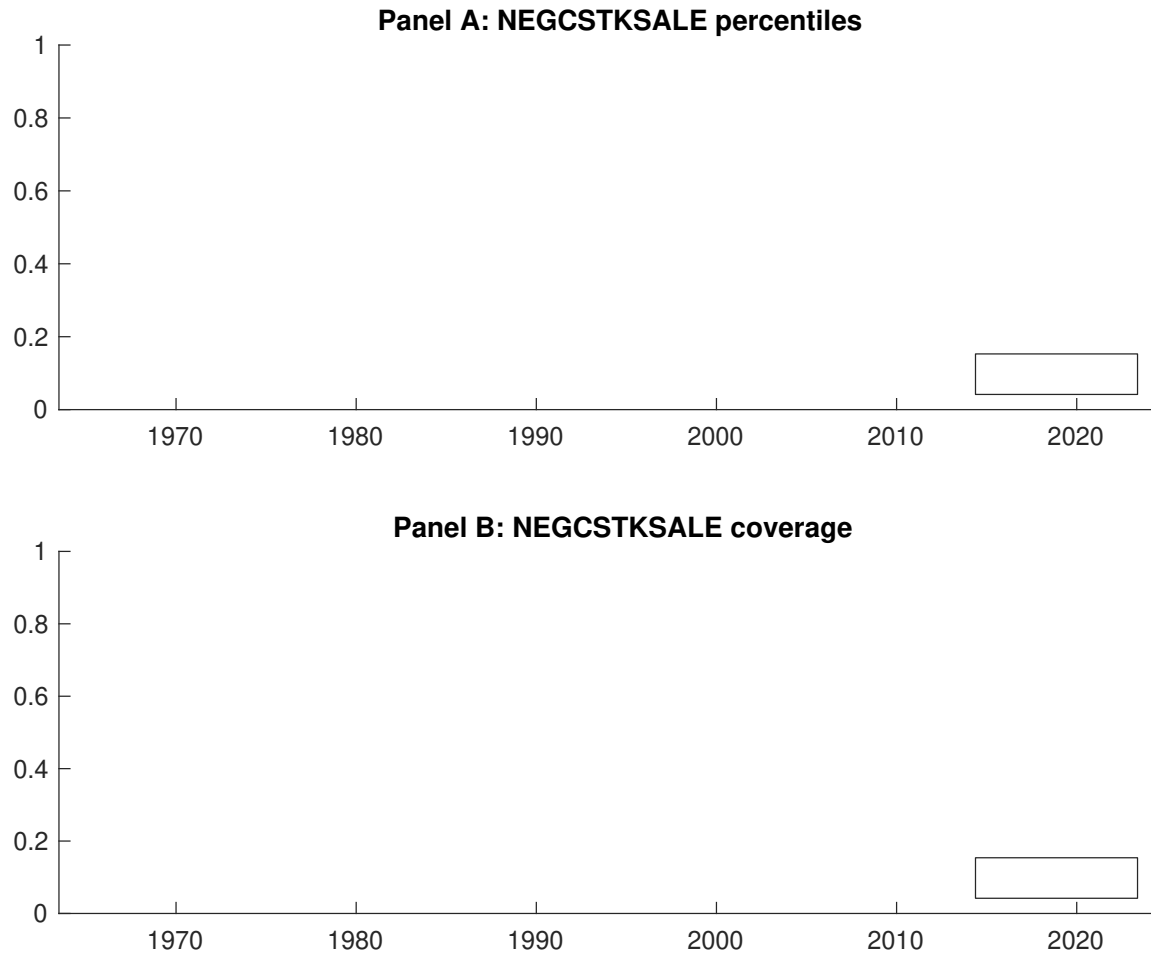


Figure 1: Times series of REE percentiles and coverage.
This figure plots descriptive statistics for REE. Panel A shows cross-sectional percentiles of REE over the sample. Panel B plots the monthly coverage of REE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on REE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on REE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47 [3.29]	0.53 [3.32]	0.58 [3.37]	0.60 [3.13]	0.75 [3.76]	0.28 [2.34]
α_{CAPM}	0.00 [0.04]	-0.01 [-0.25]	-0.02 [-0.42]	-0.06 [-1.18]	0.07 [1.20]	0.07 [0.68]
α_{FF3}	-0.08 [-1.77]	-0.06 [-1.28]	-0.02 [-0.43]	-0.07 [-1.43]	0.15 [2.81]	0.24 [2.77]
α_{FF4}	-0.09 [-1.83]	-0.08 [-1.64]	-0.02 [-0.44]	-0.00 [-0.06]	0.17 [3.12]	0.26 [3.00]
α_{FF5}	-0.14 [-2.97]	-0.17 [-3.82]	-0.03 [-0.72]	-0.01 [-0.25]	0.23 [4.26]	0.37 [4.38]
α_{FF6}	-0.14 [-2.89]	-0.17 [-3.90]	-0.03 [-0.72]	0.04 [0.82]	0.24 [4.37]	0.38 [4.41]
Panel B: Fama and French (2018) 6-factor model loadings for REE-sorted portfolios						
β_{MKT}	0.88 [75.77]	0.99 [92.21]	1.02 [105.41]	1.07 [91.38]	1.07 [80.64]	0.19 [9.03]
β_{SMB}	-0.12 [-7.02]	-0.01 [-0.67]	-0.05 [-3.44]	0.07 [4.34]	0.10 [5.42]	0.22 [7.38]
β_{HML}	0.20 [8.70]	0.06 [2.92]	0.02 [1.05]	0.07 [2.89]	-0.17 [-6.68]	-0.37 [-9.13]
β_{RMW}	0.07 [2.96]	0.21 [10.11]	0.04 [2.04]	-0.04 [-1.74]	-0.12 [-4.52]	-0.18 [-4.54]
β_{CMA}	0.18 [5.31]	0.18 [6.05]	-0.01 [-0.40]	-0.18 [-5.46]	-0.19 [-5.00]	-0.36 [-6.16]
β_{UMD}	-0.00 [-0.20]	0.01 [0.80]	0.00 [0.04]	-0.07 [-6.43]	-0.01 [-1.01]	-0.01 [-0.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	824	638	727	716	804	
me (\$10 ⁶)	2131	1839	1990	1470	2860	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the REE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28 [2.34]	0.07 [0.68]	0.24 [2.77]	0.26 [3.00]	0.37 [4.38]	0.38 [4.41]
Quintile	NYSE	EW	0.21 [2.42]	0.04 [0.55]	0.06 [0.94]	0.13 [1.97]	0.05 [0.78]	0.11 [1.67]
Quintile	Name	VW	0.26 [2.05]	0.03 [0.28]	0.20 [2.39]	0.23 [2.70]	0.33 [3.82]	0.34 [3.93]
Quintile	Cap	VW	0.28 [2.25]	0.07 [0.64]	0.24 [2.73]	0.28 [3.04]	0.37 [4.17]	0.39 [4.29]
Decile	NYSE	VW	0.41 [2.69]	0.17 [1.21]	0.40 [3.68]	0.37 [3.32]	0.49 [4.37]	0.45 [3.99]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [2.22]	0.05 [0.46]	0.20 [2.29]	0.21 [2.45]	0.29 [3.42]	0.29 [3.47]
Quintile	NYSE	EW	0.11 [1.23]			0.00 [0.02]		
Quintile	Name	VW	0.24 [1.92]	0.01 [0.05]	0.16 [1.88]	0.18 [2.09]	0.24 [2.84]	0.25 [2.93]
Quintile	Cap	VW	0.26 [2.14]	0.05 [0.44]	0.20 [2.29]	0.22 [2.49]	0.29 [3.32]	0.30 [3.42]
Decile	NYSE	VW	0.39 [2.57]	0.15 [1.06]	0.36 [3.24]	0.34 [3.08]	0.40 [3.68]	0.38 [3.50]

Table 3: Conditional sort on size and REE

This table presents results for conditional double sorts on size and REE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on REE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high REE and short stocks with low REE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	REE Quintiles					REE Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.57 [2.58]	0.77 [3.19]	0.77 [3.19]	0.83 [3.31]	0.91 [3.37]	0.33 [2.91]	0.19 [1.77]	0.18 [1.66]	0.22 [2.05]	0.13 [1.14]	0.17 [1.54]
	(2)	0.63 [3.19]	0.81 [3.75]	0.83 [3.52]	0.82 [3.46]	0.82 [3.31]	0.19 [1.69]	0.02 [0.23]	0.12 [1.27]	0.10 [1.04]	0.11 [1.18]	0.10 [1.00]
	(3)	0.58 [3.40]	0.65 [3.22]	0.77 [3.57]	0.87 [3.97]	0.85 [3.58]	0.27 [2.16]	0.05 [0.44]	0.15 [1.60]	0.13 [1.40]	0.15 [1.66]	0.14 [1.48]
	(4)	0.57 [3.49]	0.59 [3.21]	0.69 [3.44]	0.73 [3.53]	0.84 [3.86]	0.27 [2.26]	0.08 [0.71]	0.22 [2.52]	0.20 [2.28]	0.26 [2.93]	0.24 [2.67]
	(5)	0.45 [3.15]	0.51 [3.18]	0.49 [3.05]	0.54 [2.90]	0.67 [3.50]	0.23 [1.73]	0.02 [0.15]	0.19 [1.92]	0.24 [2.33]	0.36 [3.57]	0.38 [3.75]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	REE Quintiles					Average market capitalization (\$10 ⁶)						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	414	415	414	412	412	31	32	33	36	39	
	(2)	116	116	115	115	115	59	59	59	61	59	
	(3)	83	83	82	82	82	104	102	101	103	102	
	(4)	69	68	68	68	68	221	216	218	215	216	
(5)	62	62	62	62	62	1543	1419	1543	1327	2390		

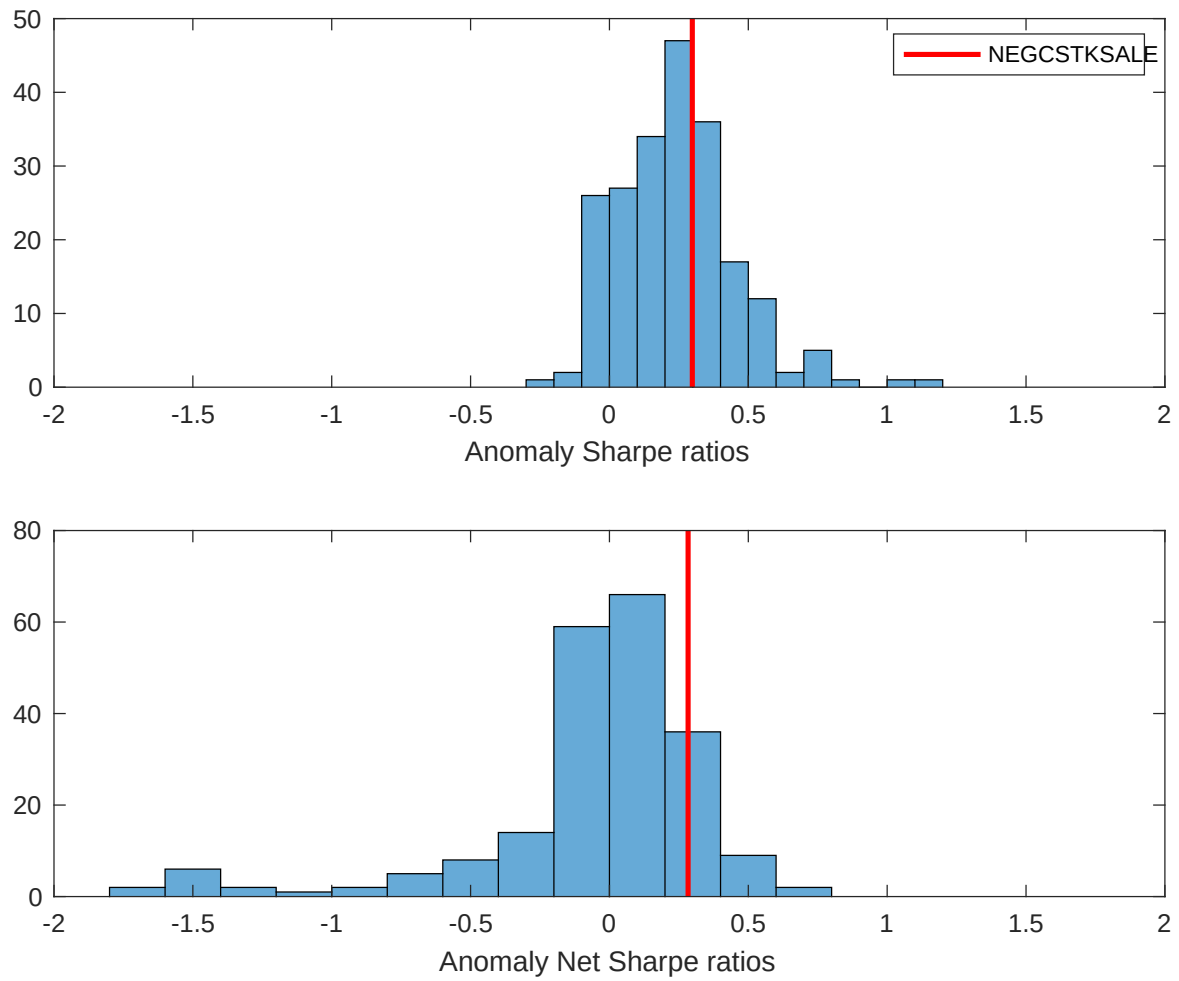


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the REE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

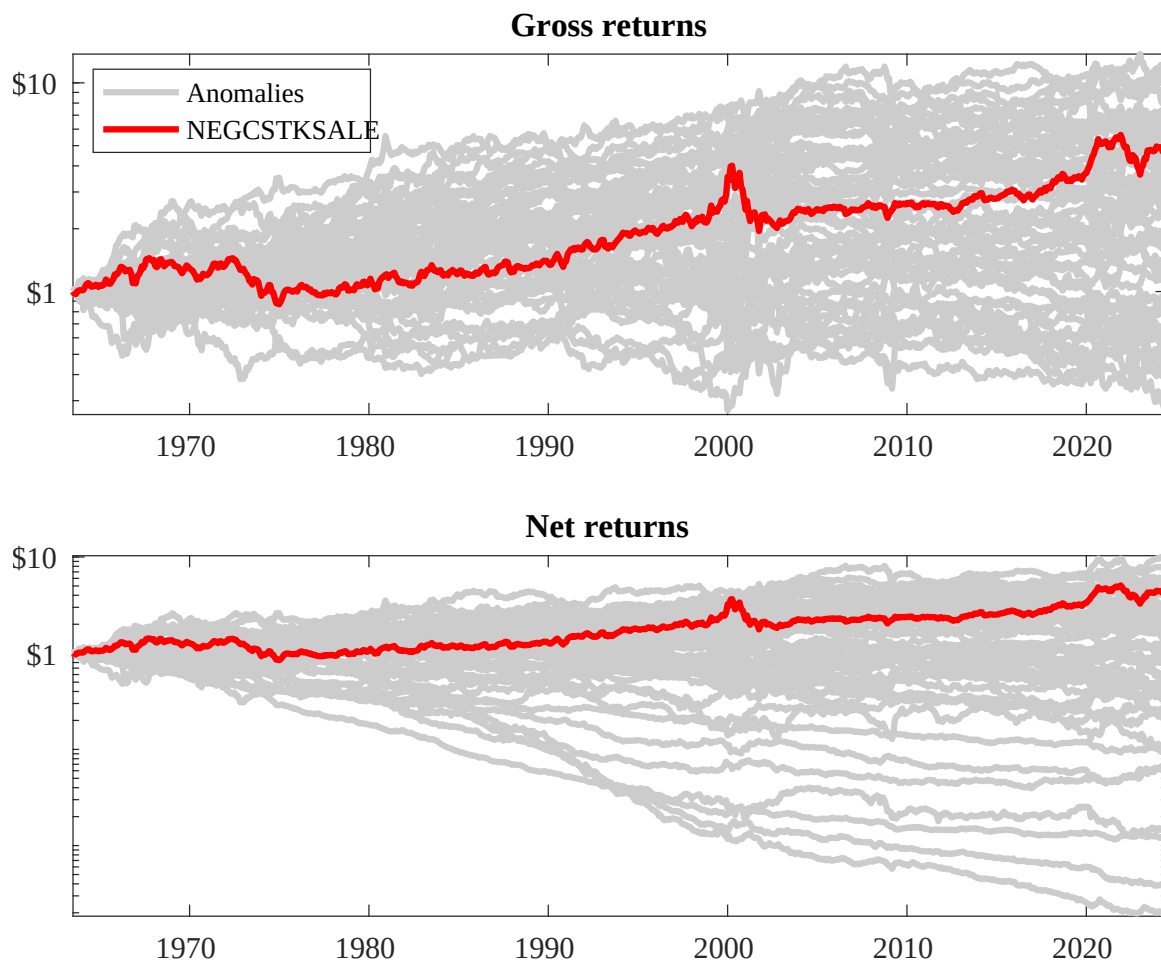
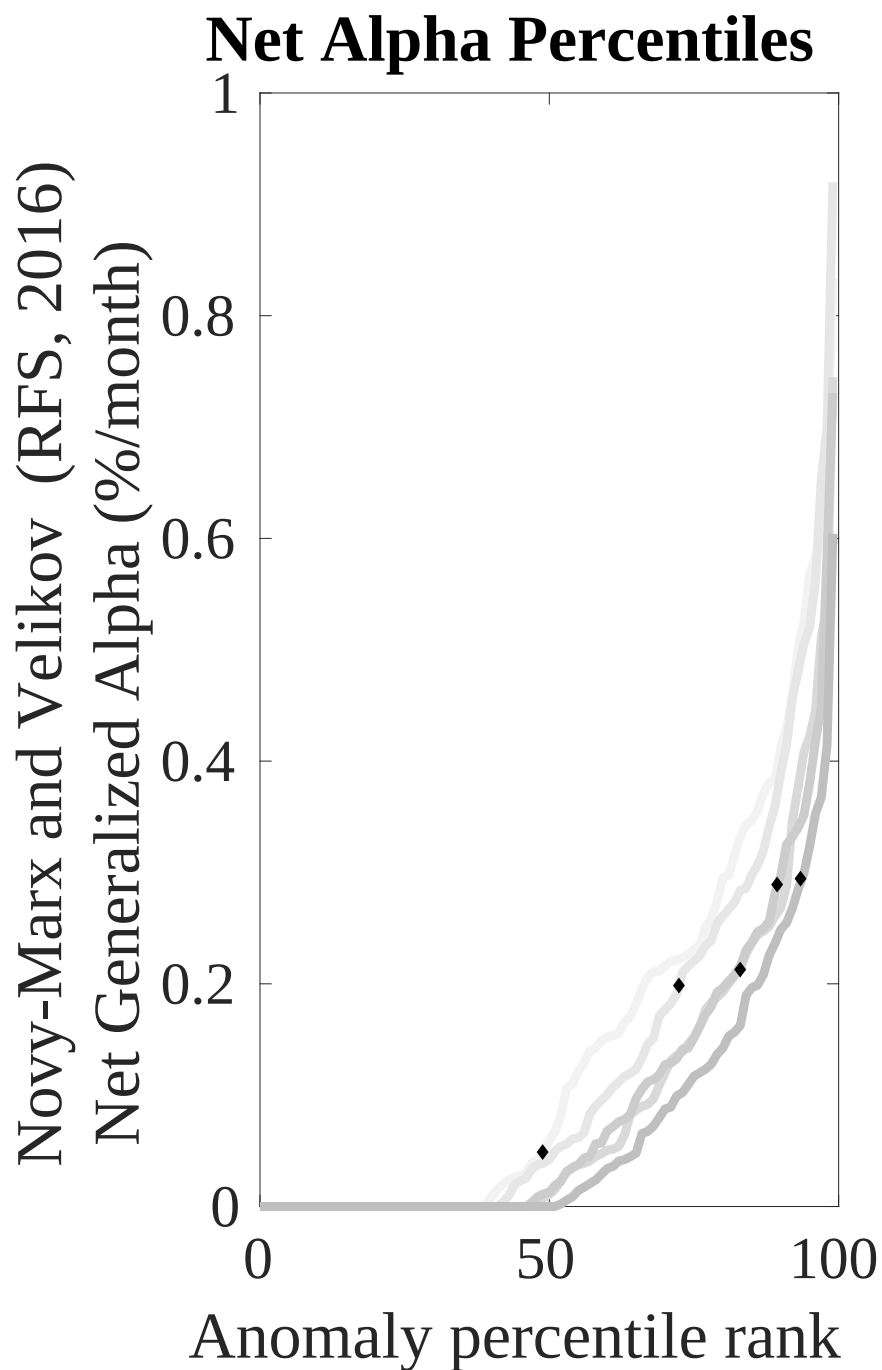
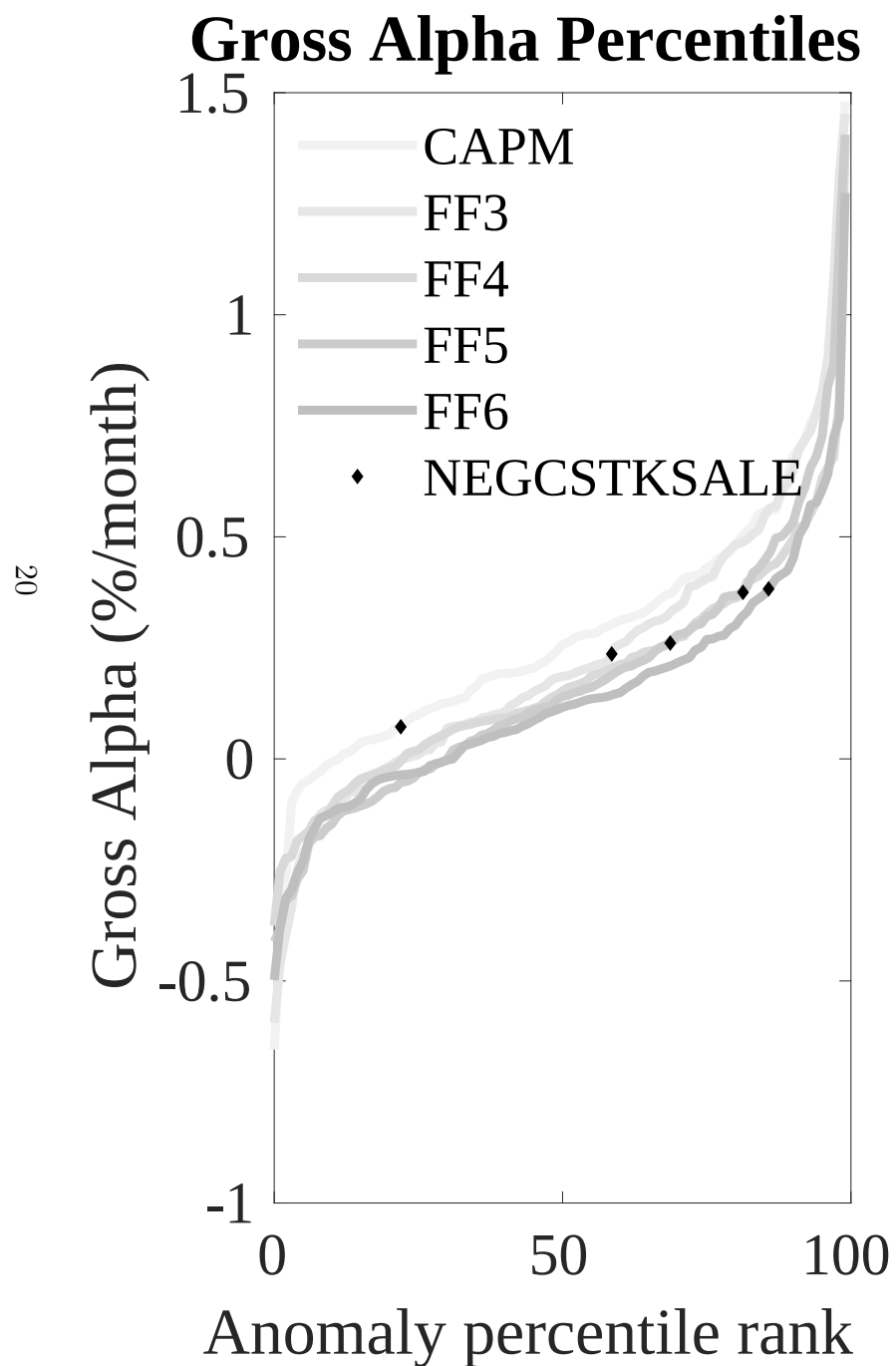


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the REE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



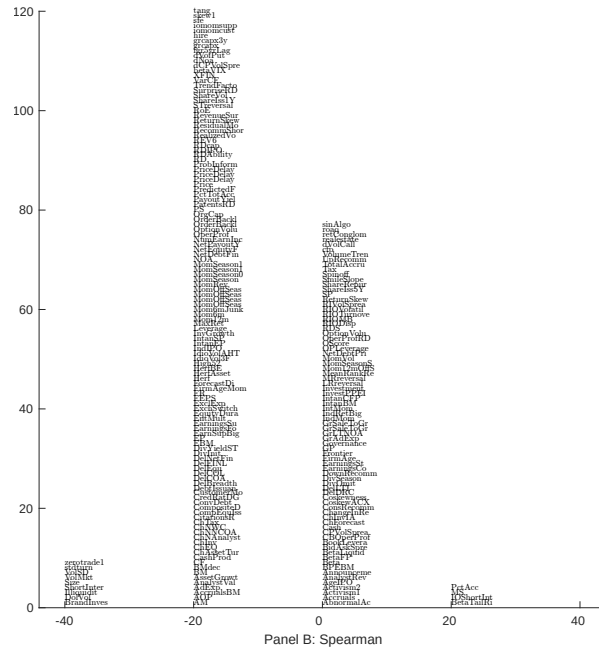
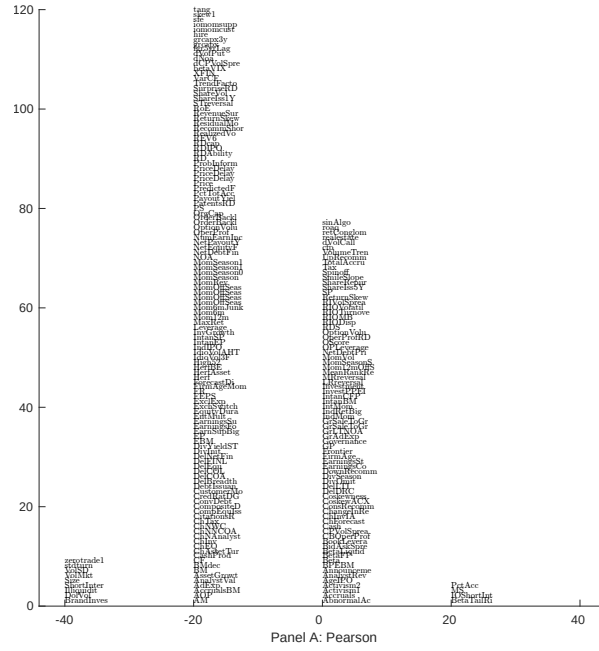


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with REE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

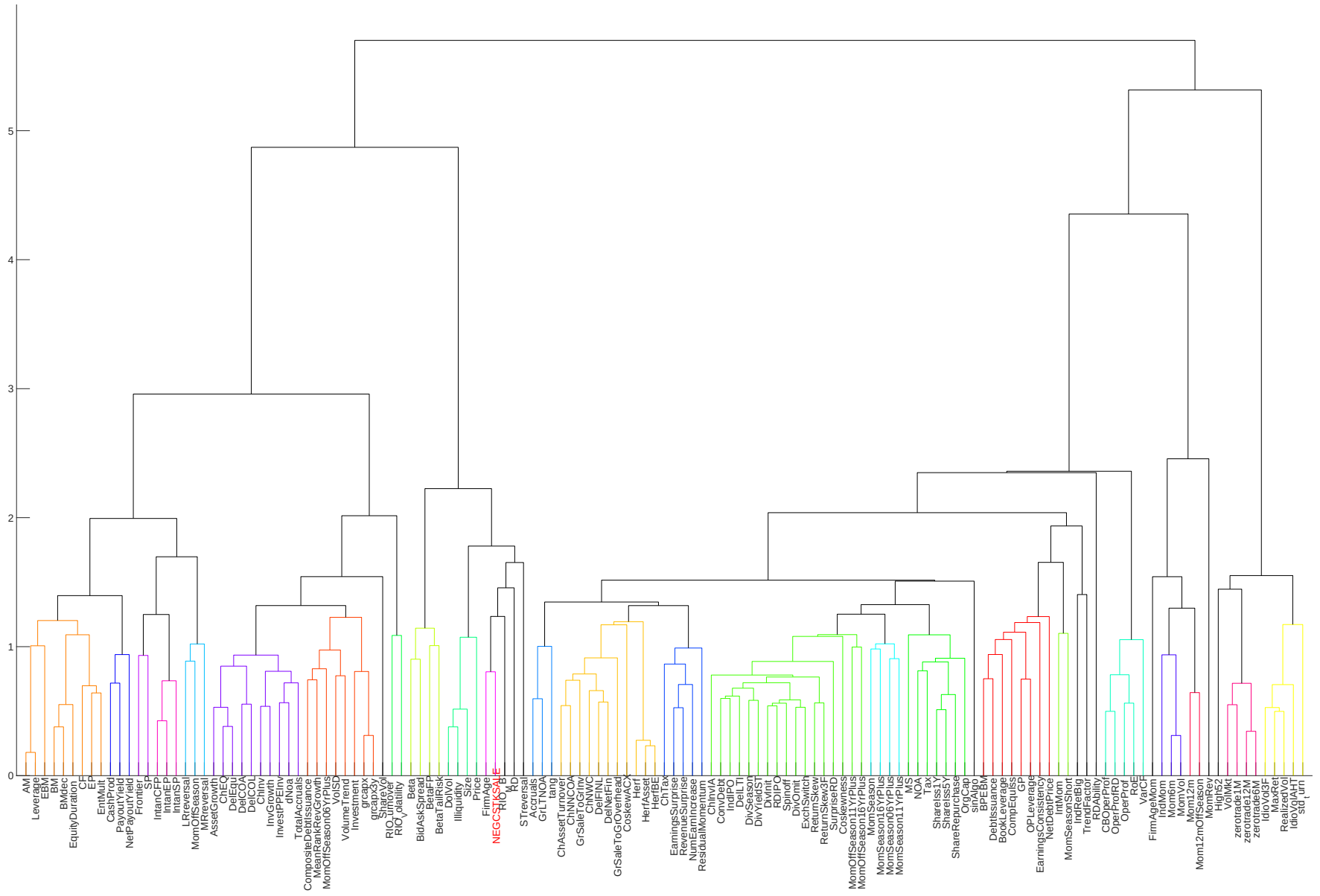


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

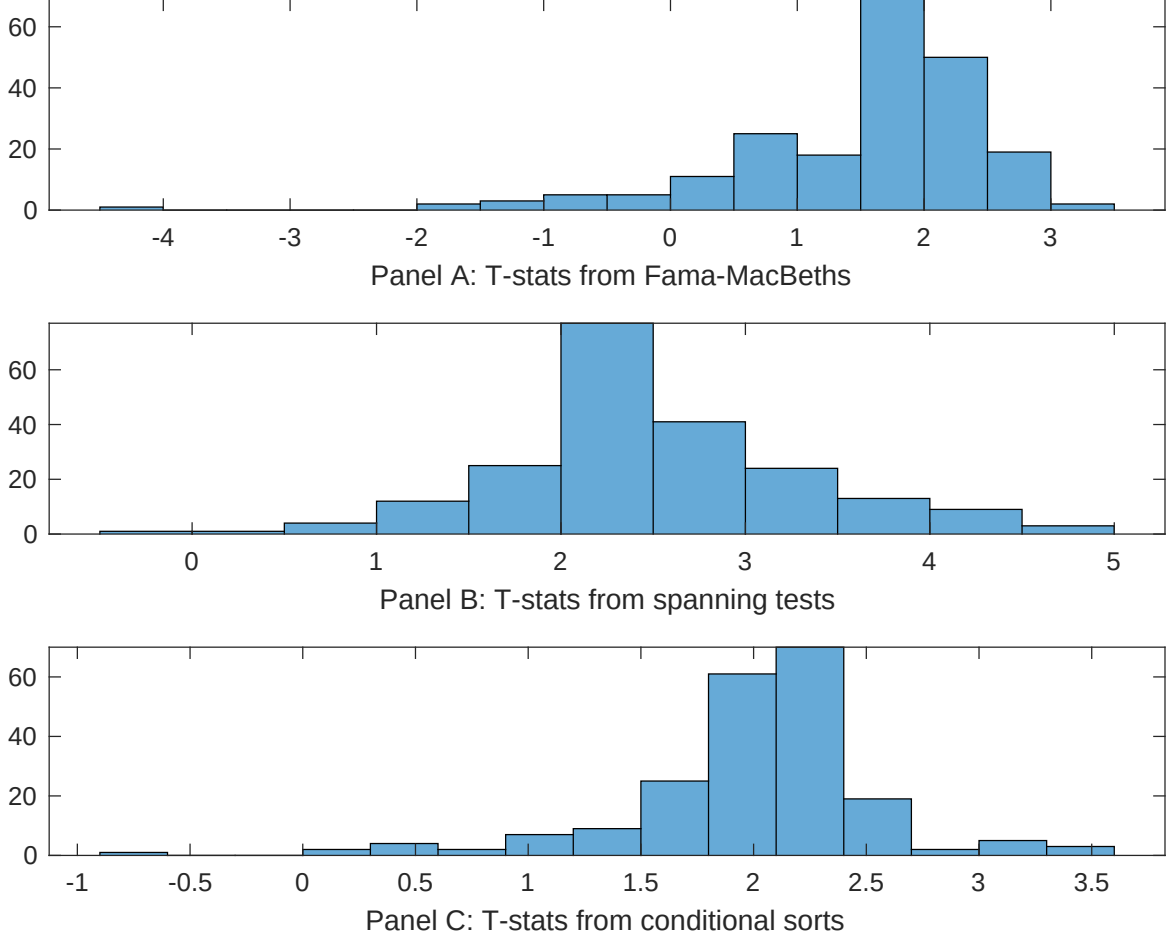


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of REE conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{REE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{REE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on REE. Stocks are finally grouped into five REE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted REE trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on REE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{REE} REE_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [7.35]	0.11 [4.82]	0.11 [4.68]	0.11 [4.71]	0.12 [5.39]	0.14 [6.46]	0.14 [6.84]
REE	0.20 [1.83]	0.20 [1.65]	0.20 [1.65]	0.19 [1.60]	0.32 [1.98]	0.17 [1.27]	0.38 [2.10]
Anomaly 1	0.21 [1.57]						0.11 [1.67]
Anomaly 2		0.19 [1.38]					0.24 [0.95]
Anomaly 3			0.19 [2.45]				-0.14 [-1.43]
Anomaly 4				0.12 [3.23]			0.94 [0.52]
Anomaly 5					0.17 [0.69]		-0.26 [-0.14]
Anomaly 6						0.95 [1.18]	0.94 [1.37]
# months	727	727	727	727	492	504	487
$\bar{R}^2(\%)$	2	1	1	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the REE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{REE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Volume to market equity, Days with zero trades, Days with zero trades, Days with zero trades, Predicted Analyst forecast error, Long-term EPS forecast. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.29 [3.58]	0.34 [4.17]	0.38 [4.63]	0.35 [4.29]	0.40 [4.25]	0.39 [4.36]	0.33 [3.82]
Anomaly 1	-28.17 [-11.19]						-3.69 [-0.60]
Anomaly 2		-26.36 [-10.67]					-1.45 [-0.23]
Anomaly 3			-23.16 [-9.19]				3.00 [0.34]
Anomaly 4				-24.46 [-9.69]			-16.47 [-2.18]
Anomaly 5					-25.36 [-8.91]		-11.15 [-3.78]
Anomaly 6						-36.92 [-12.16]	-22.07 [-6.04]
mkt	6.10 [2.72]	7.75 [3.51]	8.98 [3.98]	8.71 [3.90]	15.25 [6.63]	7.75 [3.32]	1.76 [0.73]
smb	5.62 [1.77]	12.62 [4.26]	18.71 [6.47]	21.41 [7.53]	16.99 [4.95]	5.01 [1.51]	5.50 [1.48]
hml	-28.47 [-7.55]	-29.62 [-7.83]	-30.09 [-7.80]	-30.53 [-7.98]	-24.43 [-5.49]	-20.25 [-4.77]	-16.89 [-4.10]
rmw	-3.29 [-0.82]	-2.01 [-0.49]	-3.48 [-0.83]	-4.01 [-0.98]	-13.52 [-3.03]	-13.10 [-3.15]	-2.16 [-0.49]
cma	-21.27 [-3.77]	-22.53 [-3.98]	-23.42 [-4.05]	-21.88 [-3.79]	-40.89 [-6.68]	-10.75 [-1.69]	-13.76 [-2.24]
umd	5.02 [2.54]	-1.54 [-0.80]	-0.78 [-0.40]	-0.12 [-0.06]	-1.66 [-0.78]	2.17 [1.06]	3.00 [1.30]
# months	728	728	728	728	492	504	488
$\bar{R}^2(\%)$	61	60	59	59	68	70	74

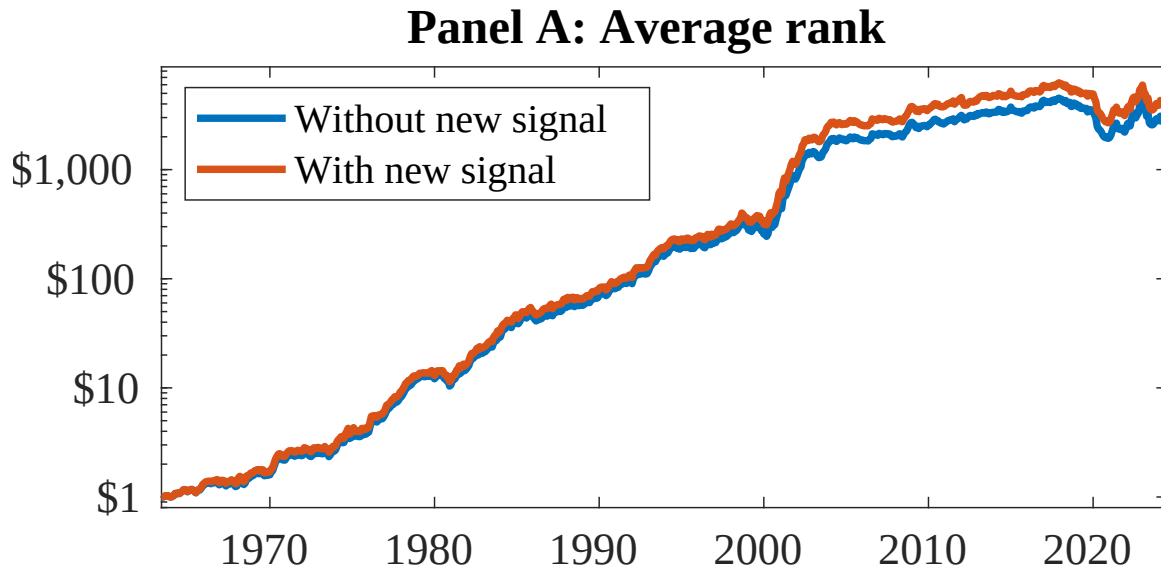


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as REE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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